

MAGeCK Count Report

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Parameters

comparison_name is the prefix of your output file, defined by the “-n” parameter in your “mageck test” command. The system will look for the following files to generate this report:

- comparison_name.countsummary.txt
- comparison_name.count_normalized.txt
- comparison_name.log

```
# define the comparison_name here; for example,
# comparison_name='demo'
comparison_name='allnonorm'
```

Note: if the sample labels are too long, they may cause issues in some figures. If this is the case, set the following variable to “TRUE”, and a numbered label will be shown instead of the original label.

```
REPLACE_LABEL=FALSE
```

Preprocessing

Reading input files. If any of these files are problematic, an error message will be shown below.

```
cstable=read.table(count_summary_file,header = T,as.is = T)
nc_table=read.table(normalized_cnt_file,header = T,as.is = T)
```

Summary

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr   1.5.1
## ✓ ggplot2    3.5.2      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.4
## — Conflicts ————— tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

The summary of the count command is as follows.

Count command summary

File	Label	Reads	Mapped	Percentage	TotalsgRNAs	Zerocounts	GiniIndex
21062D-07-03_S104_L006_R1_001.fastq	T0	26739135	15452375	0.5779	13008	462	0.1028
21062D-07-04_S105_L006_R1_001.fastq	D7	30680921	17345990	0.5654	13008	2199	0.3762
01_subset_PSU/21062D-07-01_S102_L006_R1_001.fastq	UL	26434164	14712582	0.5566	13008	3563	0.4958
21062D-07-02_S103_L006_R1_001.fastq	LL	30520874	17278471	0.5661	13008	3928	0.6153

The meanings of the columns are as follows.

- File: The filename of fastq file;
- Label: Assigned label;
- Reads: The total read count in the fastq file;
- Mapped: Reads that can be mapped to gRNA library;
- Percentage: The percentage of mapped reads;
- TotalsgRNAs: The number of sgRNAs in the library;
- ZeroCounts: The number of sgRNA with 0 read counts;
- GiniIndex: The Gini Index of the read count distribution. Gini index can be used to measure the evenness of the read counts, and a smaller value means a more even distribution of the read counts.

If -day0label and -gmt-file options are provided, the following metrics will display the degree of negative selections of essential genes (provided by -gmt-file).

Count command summary

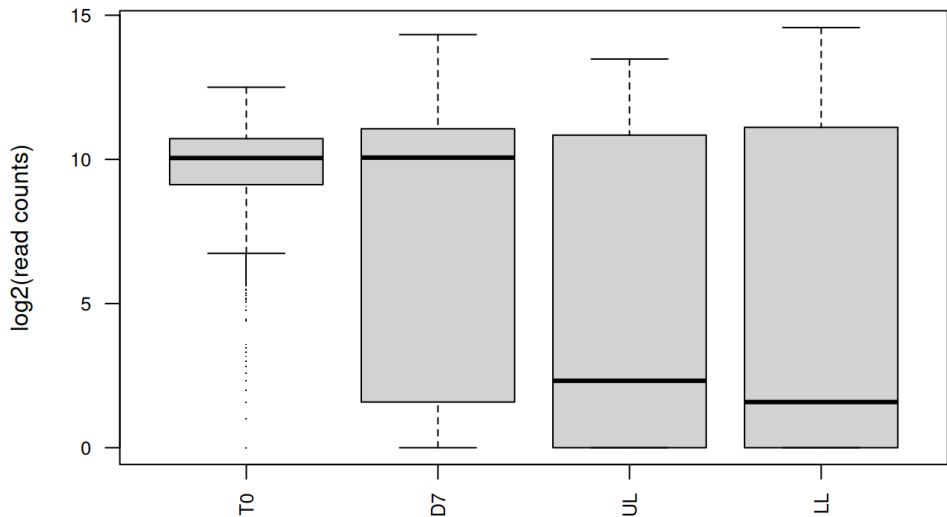
File	Label	NegSelQC	NegSelQCPval	NegSelQCPvalPermutation	NegSelQCPvalPermutationFDR	NegSelQCGene
21062D-07-03_S104_L006_R1_001.fastq	T0	0	1	1	1	1
21062D-07-04_S105_L006_R1_001.fastq	D7	0	1	1	1	1
01_subset_PSU/21062D-07-01_S102_L006_R1_001.fastq	UL	0	1	1	1	1
21062D-07-02_S103_L006_R1_001.fastq	LL	0	1	1	1	1

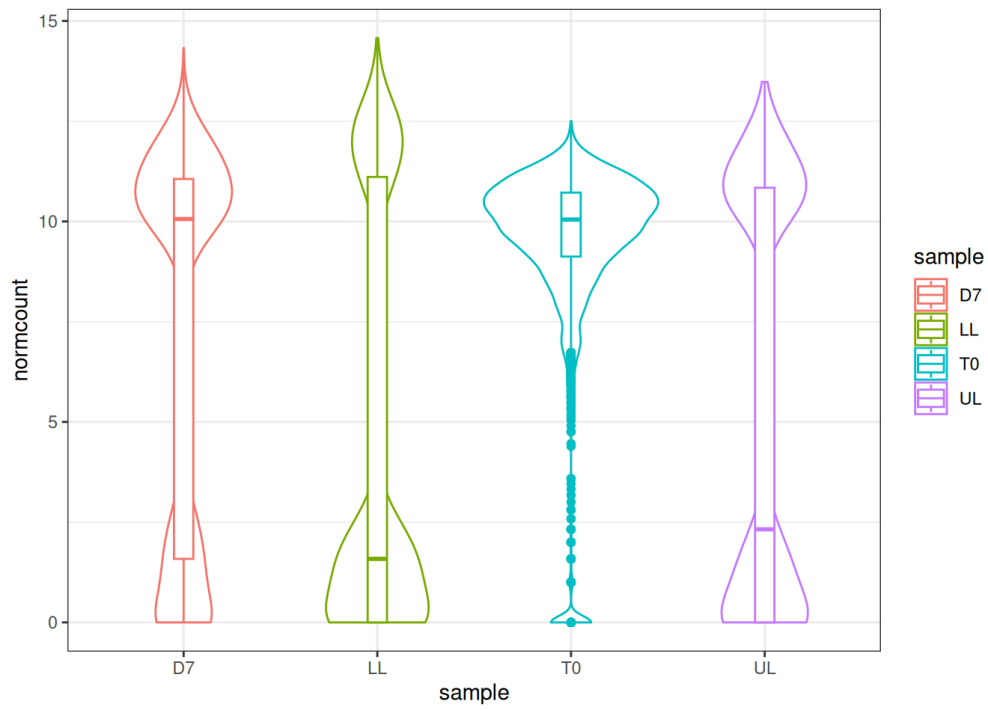
The meanings of the columns are as follows.

- NegSelQC: the enrichment score (ES) of essential genes in the negative selection list. The score is calculated using GSEA;
- NegSelQCPval: the associated p value of the enrichment score;
- NegSelQCPvalPermutation: the permuted p value of the enrichment score;
- NegSelQCPvalPermutationFDR: the adjusted permuted p value;
- NegSelQCGene: the number of genes used for the analysis.

Normalized read count distribution of all samples

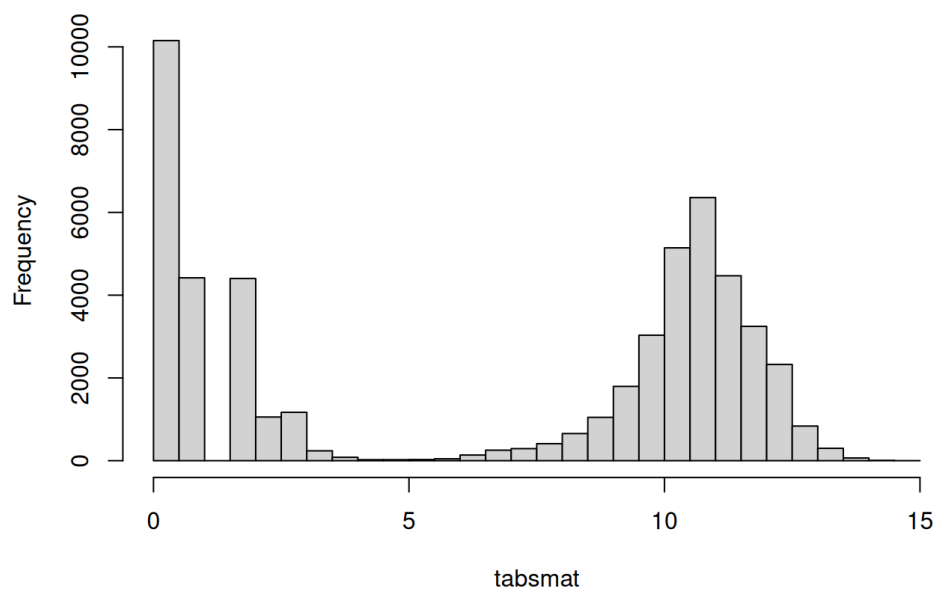
The following figure shows the distribution of median-normalized read counts in all samples.



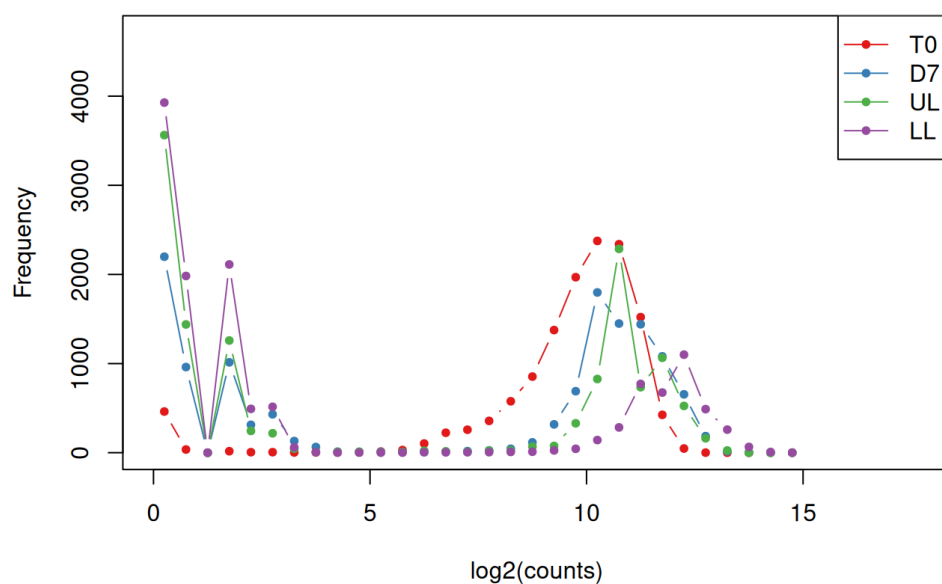


The following figure shows the histogram of median-normalized read counts in all samples.

Histogram of tabsmat

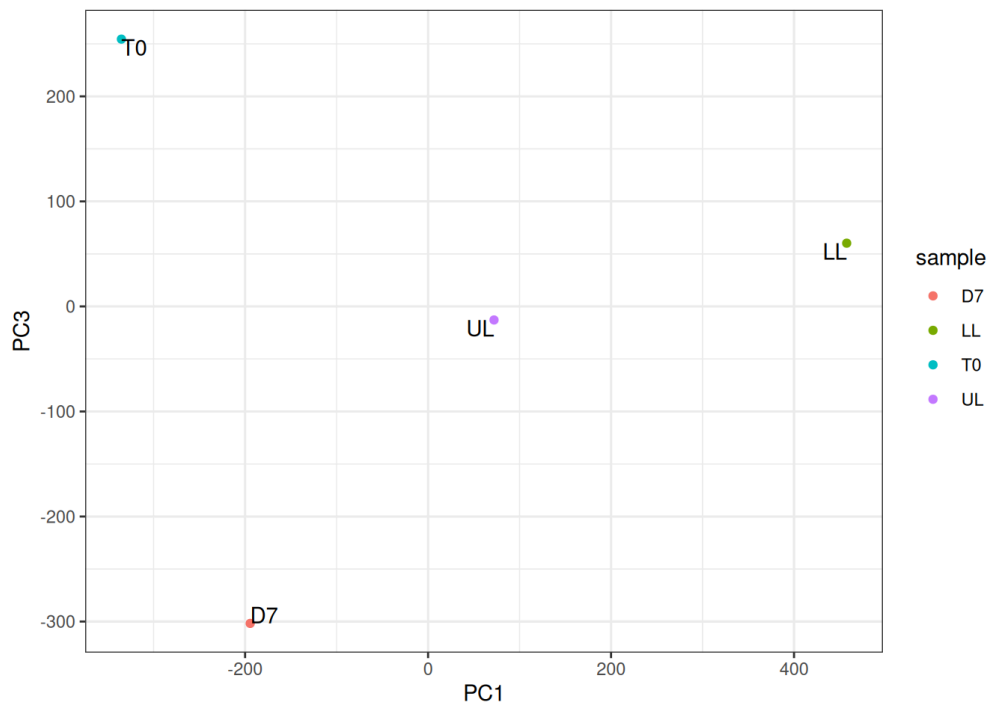
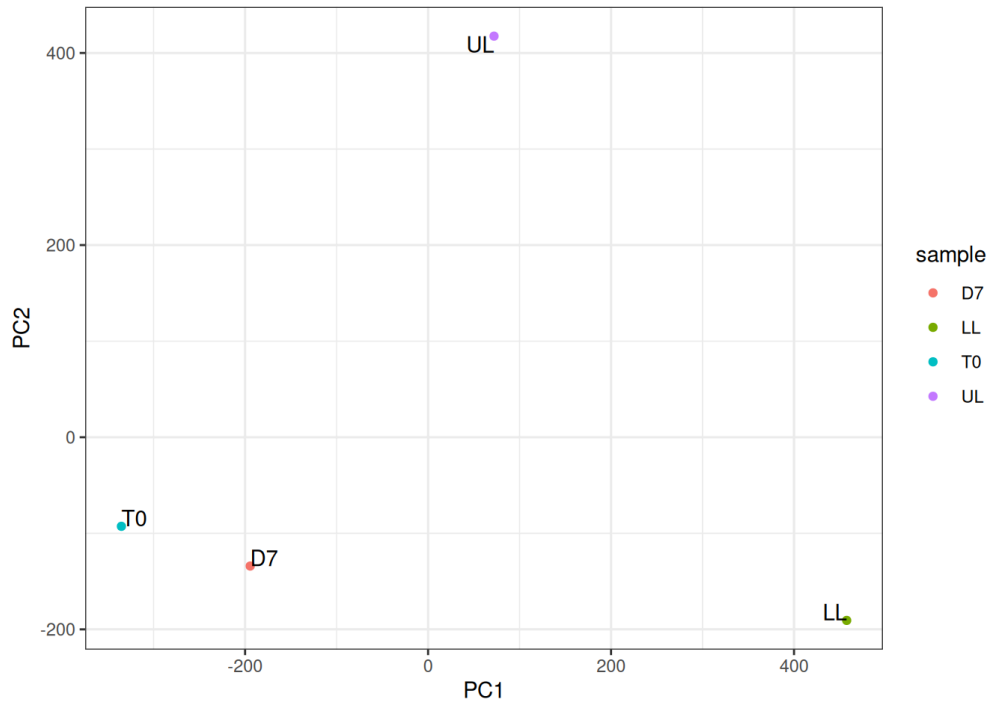


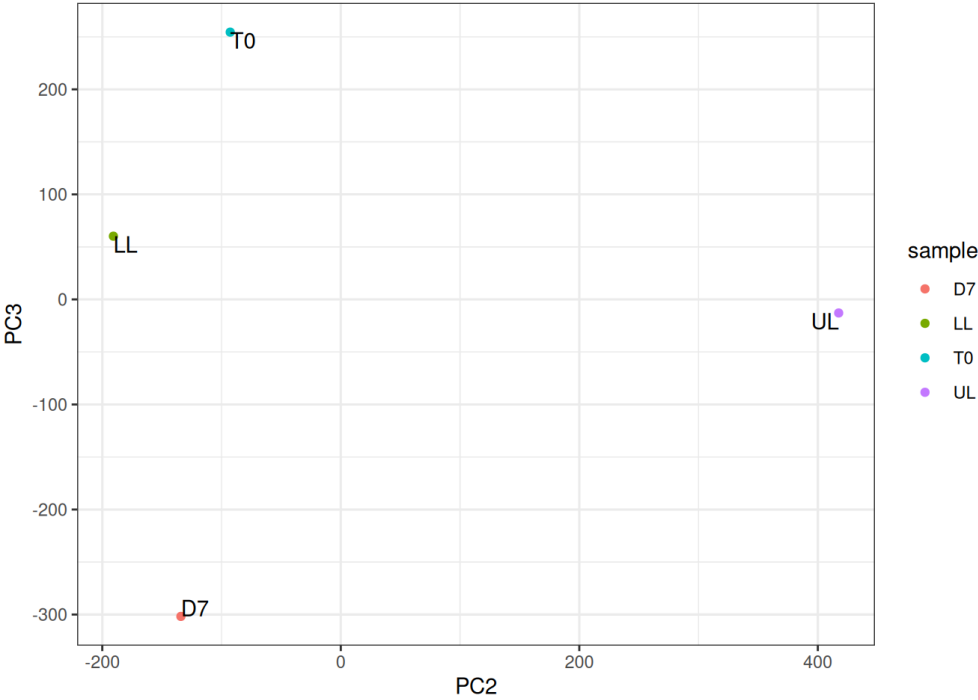
Distribution of read counts



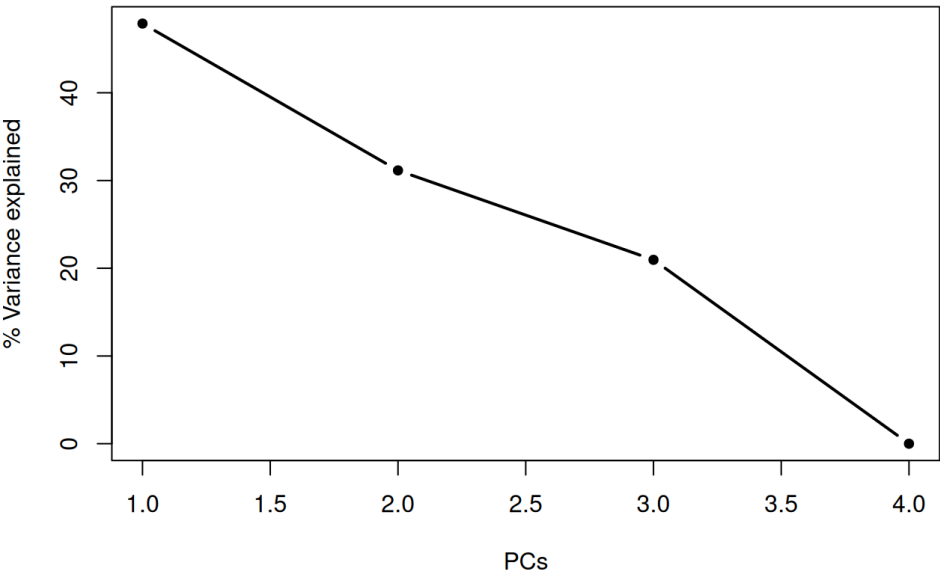
Principle Component Analysis

The following figure shows the first 2 principle components (PCs) from the Principle Component Analysis (PCA), and the percentage of variances explained by the top PCs.





The variance of the PCs



Sample clustering

The following figure shows the sample clustering result.

